Averaging Weights Leads to Wider Optima and Better Generalization

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Motivation: Standard SGD Training

Standard approach: SGD with decaying learning rate until convergence

Key Observations

- Local optima found by SGD can be connected by curves of constant loss
- There exist regions of high-performing networks, not just isolated points
- But where does SGD converge within these regions?

Core Problem

SGD converges to points on the periphery of optimal regions. Can we find more central, flatter solutions?

Loss Surface Geometry

SGD with Constant Learning Rate

Running SGD with constant learning rate α approximately samples:

$$w_i \sim \mathcal{N}(\hat{w}, \Sigma(\alpha))$$

Standard SGD:

- Decaying LR picks one point on surface
- Stops at periphery
- Near directions of sharp ascent

Key Insight:

- Train loss $\neq$ test error surfaces
- They are shifted
- Central points generalize better

Solution

Average multiple points $\rightarrow$ move to the center!

Stochastic Weight Averaging (SWA)

The Algorithm

Collect weight vectors $\{w_1, w_2, \ldots, w_n\}$ along SGD trajectory:

$$w_{\text{SWA}} = \frac{1}{n}\sum_{i=1}^{n} w_i$$

1. Train with standard SGD for $\sim$75% of budget $\rightarrow$ get $\hat{w}$
2. Continue with cyclical or constant learning rate
3. Periodically save weights and update running average:

$$w_{\text{SWA}} \leftarrow \frac{w_{\text{SWA}} \cdot n + w}{n+1}$$

4. Recompute batch norm statistics for w_{SWA}

Computational Cost

Memory: +10%   Time: +0-5%

Learning Rate Schedules

Cyclical Schedule
Linearly decrease from α_1 to α_2 in cycles:

$$\alpha(i) = (1 - t(i))\alpha_1 + t(i)\alpha_2$$

$$t(i) = \frac{1}{c}(\mathrm{mod}(i-1, c) + 1)$$

At cycle end: capture weights, jump back to α_1

Constant Schedule
Simply set $\alpha(i) = \alpha_1$ for all iterations

Capture weights every epoch

Hyperparameter Guidelines

- α_1: between max and min LR from standard training (0.01-0.05)
- Cycle length c: 3-5 epochs works well
- Method is robust to hyperparameter choices

Why Does SWA Work?

1. Geometric Interpretation

SGD with cyclical/constant LR traverses surface of optimal set
$\rightarrow$ Averaging moves inside to more central points

2. Connection to Ensembling

Consider network predictions $f(\cdot)$ and Taylor expansion:

$$f(w_i) = f(w_{\text{SWA}}) + \langle \nabla f(w_{\text{SWA}}), \Delta_i \rangle + O(\|\Delta_i\|^2)$$

where $\Delta_i = w_i - w_{\text{SWA}}$ and $\sum_i \Delta_i = 0$. Thus:

$$\bar{f} - f(w_{\text{SWA}}) = O(\Delta^2)$$

SWA approximates ensemble to second order with single model!

3. Solution Width

SWA finds demonstrably flatter solutions than SGD

Loss Surface Visualization

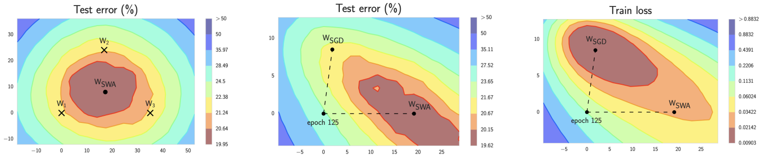

Figure: PreResNet-164 on CIFAR-100. Left: FGE samples on periphery, SWA in center. Middle & Right: SGD vs SWA on test error and train loss surfaces.

- w_{SWA}: test error 19.62%, train loss 0.00903 (center)
- w_{SGD}: test error 20.15%, train loss 0.02142 (periphery)
- SWA is more centrally located

CIFAR Results

Model	SGD	FGE	SWA 1.0B	SWA 1.25B	SWA 1.5B
			CIFAR-100		
VGG-16	72.55	74.26	73.91	74.17	**74.27**
ResNet-164	78.49	79.84	79.77	80.18	**80.35**
WRN-28-10	80.82	82.27	81.46	81.91	**82.15**
PyramidNet	83.41	—	—	83.93	**84.16**
			CIFAR-10		
VGG-16	93.25	93.52	93.59	93.70	**93.64**
ResNet-164	95.28	95.45	95.56	95.77	**95.83**
WRN-28-10	96.18	96.36	96.45	96.64	**96.79**
Shake-Shake	96.93	—	—	97.16	**97.12**

Improvements: 0.75-1.5% on CIFAR-100, 0.4-0.6% on CIFAR-10

Key result: Single SWA model $\approx$ FGE ensemble!

ImageNet Results

Using pretrained PyTorch models, SWA runs for just 10 epochs

Model	SGD	SWA (10 epochs)	Improvement
ResNet-50	76.15	**76.97 $\pm$ 0.05**	+0.82%
ResNet-152	78.31	**78.94 $\pm$ 0.07**	+0.63%
DenseNet-161	77.65	**78.44 $\pm$ 0.06**	+0.79%

Takeaway

0.6-0.9% improvement on large-scale dataset with minimal cost
Substantial given difficulty of improving strong baselines

Loss Surface Asymmetry

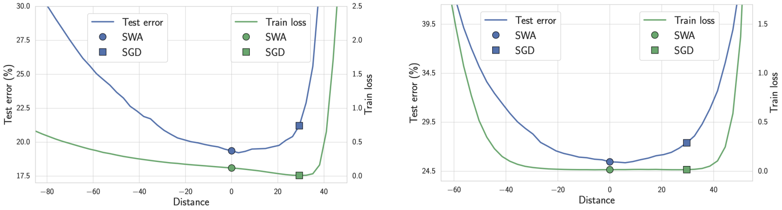

Figure: Train loss and test error along line connecting SGD and SWA (PreResNet-164, CIFAR-100)

- w_{SGD} near boundary of wide flat region
- Beyond w_{SGD} (away from SWA): steep ascent
- w_{SWA} more centrally located in flat region
- Train loss vs test error surfaces are shifted

Training from Scratch with Constant LR

Experiment: Wide ResNet-28-10 on CIFAR-100

Train for 300 epochs with constant $\alpha = 0.05$, average from epoch 140

Method	**Test Accuracy**
Individual models (any epoch)	$\sim$65% (oscillating)
Standard SGD (decaying LR)	80.82%
SWA (constant LR)	**81.7%**

Individual proposals never converge, but their average outperforms conventional training!

Key Insights

1. Geometry Matters
   - Loss surfaces contain regions of good solutions
   - SGD explores periphery; SWA finds center
   - Train loss $\neq$ test error (shifted surfaces)
2. Solution Width
   - SWA solutions are wider/flatter than SGD
   - Robust to perturbations $\rightarrow$ better generalization
   - w_{SWA}: slightly higher train loss, lower test error
3. Practical Benefits
   - Trivial to implement ($<$20 lines)
   - Negligible overhead (0-5%)
   - Universal: works with any architecture, optimizer, dataset
   - Single model with ensemble-level performance

Conclusion

Stochastic Weight Averaging

Simple technique with profound geometric insight:
Average weights along SGD trajectory with cyclical/constant LR

Key Contributions:

- 0.5-1.5% test accuracy improvement across architectures
- Single model matches/exceeds ensemble performance
- Demonstrably wider optima than standard SGD
- Essentially no computational overhead

Future Directions:

- Combination with large-batch training
- Bayesian deep learning (uncertainty quantification)
- Extension to RL, NLP domains
- Smarter averaging schemes for faster convergence

Code: `github.com/timgaripov/swa`